

Extra study and consolidation in Algebra

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Contents

Problem. Let

$$R = \frac{k[x, y]}{(xy)}$$

and write $\bar{x}, \bar{y}$ for the residue classes.

What is $(\bar{x})$ as an ideal of R ?

Compute

$$R/(\bar{x}), \quad R/(\bar{y}), \quad R/(\bar{x}, \bar{y}).$$

Then classify them as prime/maximal/neither.

Sol. We have $\bar{x} = x + (xy)$.

The ideal generated by $\bar{x}$ is

$$(\bar{x}) = \{\overline{xf} \mid f \in k[x, y]\}.$$

Since $(xy) \subseteq (x)$ in $k[x, y]$ we have

$$\frac{R}{(\bar{x})} = \frac{k[x, y]/(xy)}{(x)/(xy)}.$$

By the third isomorphism theorem,

$$\frac{R}{(\bar{x})} \cong \frac{k[x, y]}{(x)} \cong k[y].$$

Similarly,

$$\frac{R}{(\bar{y})} \cong \frac{k[x, y]}{(y)} \cong k[x].$$

The ideal generated by $(\bar{x}, \bar{y})$ is

$$(\bar{x}, \bar{y}) = \{\overline{xf + yg} \mid f, g \in k[x, y]\}.$$

Suppose $h \in (xy)$.

Then

$$h = \eta xy, \quad \eta \in k[x, y].$$

Notice that

$$h = \eta xy = \eta \cdot x \cdot y + 0 \cdot y \in (x, y).$$

So as ideals of $k[x, y]$,

$$(xy) \subseteq (x, y).$$

Then

$$\frac{(x, y)}{(xy)}$$

is an ideal of $R = k[x, y]/(xy)$. and

$$\frac{R}{(\bar{x}, \bar{y})} \cong \frac{k[x, y]/(xy)}{(x, y)/(xy)} \cong \frac{k[x, y]}{(x, y)} \cong k.$$

Since k is a field we have that $(\bar{x}, \bar{y})$ is maximal.

For the other two we note that polynomial rings over a field k are integral domains, so both $(\bar{x}), (\bar{y})$ are prime, but not maximal since $k[x]$ and $k[y]$ are not fields.

□